

10.2 Practice - Operations on Functions

Perform the indicated operations.

1) $g(a) = a^3 + 5a^2$
 $f(a) = 2a + 4$
Find $g(3) + f(3)$

3) $g(a) = 3a + 3$
 $f(a) = 2a - 2$
Find $(g + f)(9)$

5) $g(x) = x + 3$
 $f(x) = -x + 4$
Find $(g - f)(3)$

7) $g(x) = x^2 + 2$
 $f(x) = 2x + 5$
Find $(g - f)(0)$

9) $g(t) = t - 3$
 $h(t) = -3t^3 + 6t$
Find $g(1) + h(1)$

11) $h(t) = t + 5$
 $g(t) = 3t - 5$
Find $(h \cdot g)(5)$

13) $h(n) = 2n - 1$
 $g(n) = 3n - 5$
Find $h(0) \div g(0)$

15) $f(a) = -2a - 4$
 $g(a) = a^2 + 3$
Find $(\frac{f}{g})(7)$

17) $g(x) = -x^3 - 2$
 $h(x) = 4x$
Find $(g - h)(x)$

19) $f(x) = -3x + 2$
 $g(x) = x^2 + 5x$
Find $(f - g)(x)$

21) $g(x) = 4x + 5$
 $h(x) = x^2 + 5x$
Find $g(x) \cdot h(x)$

2) $f(x) = -3x^2 + 3x$
 $g(x) = 2x + 5$
Find $f(-4) \div g(-4)$

4) $g(x) = 4x + 3$
 $h(x) = x^3 - 2x^2$
Find $(g - h)(-1)$

6) $g(x) = -4x + 1$
 $h(x) = -2x - 1$
Find $g(5) + h(5)$

8) $g(x) = 3x + 1$
 $f(x) = x^3 + 3x^2$
Find $g(2) \cdot f(2)$

10) $f(n) = n - 5$
 $g(n) = 4n + 2$
Find $(f + g)(-8)$

12) $g(a) = 3a - 2$
 $h(a) = 4a - 2$
Find $(g + h)(-10)$

14) $g(x) = x^2 - 2$
 $h(x) = 2x + 5$
Find $g(-6) + h(-6)$

16) $g(n) = n^2 - 3$
 $h(n) = 2n - 3$
Find $(g - h)(n)$

18) $g(x) = 2x - 3$
 $h(x) = x^3 - 2x^2 + 2x$
Find $(g - h)(x)$

20) $g(t) = t - 4$
 $h(t) = 2t$
Find $(g \cdot h)(t)$

22) $g(t) = -2t^2 - 5t$
 $h(t) = t + 5$
Find $g(t) \cdot h(t)$

- 23) $f(x) = x^2 - 5x$
 $g(x) = x + 5$
Find $(f + g)(x)$
- 24) $f(x) = 4x - 4$
 $g(x) = 3x^2 - 5$
Find $(f + g)(x)$
- 25) $g(n) = n^2 + 5$
 $f(n) = 3n + 5$
Find $g(n) \div f(n)$
- 26) $f(x) = 2x + 4$
 $g(x) = 4x - 5$
Find $f(x) - g(x)$
- 27) $g(a) = -2a + 5$
 $f(a) = 3a + 5$
Find $(\frac{g}{f})(a)$
- 28) $g(t) = t^3 + 3t^2$
 $h(t) = 3t - 5$
Find $g(t) - h(t)$
- 29) $h(n) = n^3 + 4n$
 $g(n) = 4n + 5$
Find $h(n) + g(n)$
- 30) $f(x) = 4x + 2$
 $g(x) = x^2 + 2x$
Find $f(x) \div g(x)$
- 31) $g(n) = n^2 - 4n$
 $h(n) = n - 5$
Find $g(n^2) \cdot h(n^2)$
- 32) $g(n) = n + 5$
 $h(n) = 2n - 5$
Find $(g \cdot h)(-3n)$
- 33) $f(x) = 2x$
 $g(x) = -3x - 1$
Find $(f + g)(-4 - x)$
- 34) $g(a) = -2a$
 $h(a) = 3a$
Find $g(4n) \div h(4n)$
- 35) $f(t) = t^2 + 4t$
 $g(t) = 4t + 2$
Find $f(t^2) + g(t^2)$
- 36) $h(n) = 3n - 2$
 $g(n) = -3n^2 - 4n$
Find $h(\frac{n}{3}) \div g(\frac{n}{3})$
- 37) $g(a) = a^3 + 2a$
 $h(a) = 3a + 4$
Find $(\frac{g}{h})(-x)$
- 38) $g(x) = -4x + 2$
 $h(x) = x^2 - 5$
Find $g(x^2) + h(x^2)$
- 39) $f(n) = -3n^2 + 1$
 $g(n) = 2n + 1$
Find $(f - g)(\frac{n}{3})$
- 40) $f(n) = 3n + 4$
 $g(n) = n^3 - 5n$
Find $f(\frac{n}{2}) - g(\frac{n}{2})$
- 41) $f(x) = -4x + 1$
 $g(x) = 4x + 3$
Find $(f \circ g)(9)$
- 42) $g(x) = x - 1$
Find $(g \circ g)(7)$
- 43) $h(a) = 3a + 3$
 $g(a) = a + 1$
Find $(h \circ g)(5)$
- 44) $g(t) = t + 3$
 $h(t) = 2t - 5$
Find $(g \circ h)(3)$
- 45) $g(x) = x + 4$
 $h(x) = x^2 - 1$
Find $(g \circ h)(10)$
- 46) $f(a) = 2a - 4$
 $g(a) = a^2 + 2a$
Find $(f \circ g)(-4)$

$$47) \begin{aligned} f(n) &= -4n + 2 \\ g(n) &= n + 4 \\ \text{Find } (f \circ g)(9) \end{aligned}$$

$$49) \begin{aligned} g(x) &= 2x - 4 \\ h(x) &= 2x^3 + 4x^2 \\ \text{Find } (g \circ h)(3) \end{aligned}$$

$$51) \begin{aligned} g(x) &= x^2 - 5x \\ h(x) &= 4x + 4 \\ \text{Find } (g \circ h)(x) \end{aligned}$$

$$53) \begin{aligned} f(a) &= -2a + 2 \\ g(a) &= 4a \\ \text{Find } (f \circ g)(a) \end{aligned}$$

$$55) \begin{aligned} g(x) &= 4x + 4 \\ f(x) &= x^3 - 1 \\ \text{Find } (g \circ f)(x) \end{aligned}$$

$$57) \begin{aligned} g(x) &= -x + 5 \\ f(x) &= 2x - 3 \\ \text{Find } (g \circ f)(x) \end{aligned}$$

$$59) \begin{aligned} f(t) &= 4t + 3 \\ g(t) &= -4t - 2 \\ \text{Find } (f \circ g)(t) \end{aligned}$$

$$48) \begin{aligned} g(x) &= 3x + 4 \\ h(x) &= x^3 + 3x \\ \text{Find } (g \circ h)(3) \end{aligned}$$

$$50) \begin{aligned} g(a) &= a^2 + 3 \\ \text{Find } (g \circ g)(-3) \end{aligned}$$

$$52) \begin{aligned} g(a) &= 2a + 4 \\ h(a) &= -4a + 5 \\ \text{Find } (g \circ h)(a) \end{aligned}$$

$$54) \begin{aligned} g(t) &= -t - 4 \\ \text{Find } (g \circ g)(t) \end{aligned}$$

$$56) \begin{aligned} f(n) &= -2n^2 - 4n \\ g(n) &= n + 2 \\ \text{Find } (f \circ g)(n) \end{aligned}$$

$$58) \begin{aligned} g(t) &= t^3 - t \\ f(t) &= 3t - 4 \\ \text{Find } (g \circ f)(t) \end{aligned}$$

$$60) \begin{aligned} f(x) &= 3x - 4 \\ g(x) &= x^3 + 2x^2 \\ \text{Find } (f \circ g)(x) \end{aligned}$$